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$$I = \int \frac{1}{\cos x - \sin x + 1} dx$$

Stel $t = \tan\left(\frac{x}{2}\right)$

$$\sin x = \frac{2t}{1+t^2}$$

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$dx = \frac{2}{1+t^2} dt$$

Dus:

$$I = \int \frac{1}{\frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2} + 1} \cdot \frac{2}{1+t^2} dt$$

$$= \int \frac{1}{\frac{1-t^2-2t+1+t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt = \int \frac{1+t^2}{2-2t} \cdot \frac{2}{1+t^2} dt$$

$$= \int \frac{1}{1-t} dt$$

$$= -\ln|1-t| + c$$

Terug substitueren:

$$I = -\ln\left|1 - \tan\left(\frac{x}{2}\right)\right| + c$$

References